Ticket Sales and Redemption Patterns at the Toronto Island Ferry

A historical analysis of 15-minute ticket data, May 2015 to December 2025

Data: Toronto Island Ferry Tickets.csv (261,538 records) | Prepared 18 October 2026 | Companion dashboard: *streamlit run app.py*

Abstract

This report analyses 261,538 fifteen-minute records of ferry ticket sales and redemptions for Toronto Island Park, covering 2015-05-01 to 2025-12-21. The file records 12,785,293 redemptions and 12,972,051 sales. Demand is strongly seasonal and strongly concentrated in time: June to August carries 62.6% of all redemptions and December to February only 4.0%. Within a day, the busiest clock hour is 12:00-13:00 (mean 390 redemptions per hour) and the three hours from 11:00 to 14:00 carry 35% of daily redemptions. Weekends average 1.88 times weekday redemptions overall and 1.91 times in summer, but show no weekend premium in winter (0.99). Two years, 2017 and 2020, depart sharply from the pattern. All counts are tickets, not passengers; the data contain no route, capacity, queue or weather information, so the recommendations are hypotheses for operational validation rather than conclusions.

1. Business context and problem statement

Toronto Island ferries connect the Jack Layton Ferry Terminal with Centre Island, Hanlan's Point and Ward's Island. Ticket sales and redemptions are logged in 15-minute intervals. Operations teams and policy planners need to know when demand peaks, how strongly it varies by season and weekday, and how reliable the underlying records are, so that staffing, scheduling and congestion management can rest on evidence.

Interpretation limit. A ticket sale or redemption is not a passenger arrival or departure. Tickets can be bought ahead of use, redeemed later, or not redeemed. The difference between sales and redemptions is therefore reported as a *ticket-count difference*, never as net passenger movement.

2. Research objectives

- Verify the structure, coverage and quality of the supplied dataset and document every cleaning decision.
- Quantify demand by hour, weekday, month and season, and identify peak and off-peak periods.
- Compare ticket sales with redemptions where both exist, without over-interpreting the gap.
- Identify unusual spikes and anomalous years without discarding legitimate peak demand.
- Translate verified findings into operational implications, clearly separating facts from hypotheses.

3. Dataset description and provenance

The analysed file is **Toronto Island Ferry Tickets.csv** (8.6 MB, CSV) supplied with the project brief. It has four columns, all fully populated:

Column	Type	Meaning (as observed)
_id	integer	Unique row identifier, numbered newest-first (id 1 is the latest interval)
Timestamp	ISO text, no time zone	Start of a 15-minute interval, minutes only 00, 15, 30 or 45
Redemption Count	integer, 0 to 7,216	Tickets redeemed in the interval

Column	Type	Meaning (as observed)
Sales Count	integer, 0 to 7,229	Tickets sold in the interval (present in this file; the brief listed it as possibly absent)

Each row is an **interval-level aggregate**, not an individual event and not a cumulative total. The file contains 261,538 rows, one per interval with activity; 3,882 of 3,888 calendar days have data. The latest record is 2025-12-21 22:30, about 300 days before this report, so the data support **historical analysis only**; nothing in the file is a live feed. The file does not state its publisher, licence or extraction date, so provenance beyond the brief is undocumented and should be recorded by the data owner.

4. Data-cleaning methodology

The pipeline (*data_processing.py*) never modifies the source file and records every action in a cleaning report. Rules, in order:

- Timestamps are parsed; unparseable values are dropped and counted. They are kept as timezone-naive local wall-clock time and never reinterpreted as UTC. If an explicit UTC offset were present it would be converted to Toronto time.
- Counts must be numeric, non-negative integers. Missing, non-numeric, negative or fractional counts are dropped and counted; nothing is zero-filled silently.
- Rows identical in timestamp and counts are dropped as duplicates. Rows that repeat a timestamp with different counts are kept and summed per timestamp, so no tickets are lost or double-counted.
- Rows are sorted chronologically. Interval alignment, gaps, missing calendar days and DST transition days are measured.
- Outliers are flagged with a modified z-score of log(1 + count) within each (month, hour) group (threshold 3.5, minimum count 100) and retained.

Result on the supplied file: all 261,538 rows passed validation. There were no unparseable timestamps, missing values, negative or fractional counts, duplicate rows, duplicate timestamps or duplicate _id values; every timestamp lies on a 15-minute boundary. The only change was re-sorting into time order. 472 intervals (0.18%) are flagged as outliers but kept.

Absent intervals. No row has both counts equal to zero, although 27,830 rows have zero redemptions and 20,548 have zero sales. This indicates the extract omits intervals with no activity. Between each day's first and last record, 63,951 of 325,489 intervals are absent (completeness 80.3%); measured against all 96 slots on every calendar day it is 70.1%. Absent intervals are therefore reported as *absent*, and treated as zero only where stated (rolling averages). Six calendar days have no records at all: 2015-05-02, 2015-05-03, 2018-01-04, 2021-02-21, 2021-02-22, 2025-01-26.

5. Feature-engineering methodology

From each timestamp the pipeline derives: date, year, quarter, month, year-month, day of week, weekend flag (Saturday and Sunday), hour, 15-minute interval of day, and hourly start. **Season** uses the meteorological definition: Winter = December to February, Spring = March to May, Summer = June to August, Fall = September to November. Daylight-saving transition days are flagged using the America/Toronto rules. The file contains 21 such days (10 spring-forward, 11 fall-back); it has 0 records in the non-existent 02:00 hour on spring-forward days and no repeated timestamps on fall-back days, consistent with local wall-clock time. Because fall-back days cannot show the repeated hour, the 01:00 hour on those days may combine two real hours; this affects at most 11 one-hour periods in the whole file and does not change any conclusion.

6. Analytical methods and KPI definitions

All hourly, daily and monthly figures are **sums** of interval counts. Means are taken over those sums, never over averages of averages.

KPI	Definition	Interpretation
Interval / hourly / daily / monthly total	Sum of counts in the period	Volume of tickets; not passengers
Per observed interval / hour	Mean over rows (or hourly sums) present in the file	Excludes absent intervals, so it overstates a true time-average
Peak hour	Clock hour with the highest mean hourly total	12:00-13:00
Off-peak hour	Lowest mean hourly total among regular-operation hours (activity on at least 90% of days)	23:00-24:00; overnight hours are excluded because activity there is sporadic
Peak window	3-hour block of clock hours with the largest summed mean hourly totals	Share of the average daily total
Weekend / weekday ratio	Mean daily total on Sat+Sun divided by mean on Mon-Fri	>1 means weekends are busier
Seasonal Utilization Index	100 x season mean daily total / mean daily total of the whole selection	Baseline 100 = overall average day; Summer 240 means 2.4 times the average day
Low-to-peak season ratio	Mean daily total of lowest season / that of highest season	Depth of the off-season trough
Completeness	Observed intervals / intervals between each day's first and last record	Does not penalise whole days or the night
Rolling average	Trailing mean of interval counts: 1-hour = 4, 4-hour = 16 intervals, on a regular grid	Smooths 15-minute noise; absent intervals are taken as zero (stated assumption)
Sales-to-redemption ratio	Total sales / total redemptions	Ticket-count comparison, not passenger flow

No hypothesis tests or confidence intervals are reported: the analysis is descriptive, and statements of difference refer to the observed means in this file. No external sources were consulted; every figure derives from the supplied dataset.

7. Exploratory data analysis and findings

Labels: **[Observed]** a calculated fact; **[Hypothesis]** a plausible explanation the data cannot confirm; **[Recommendation]** an action needing operational validation.

7.1 Coverage and the long-run picture

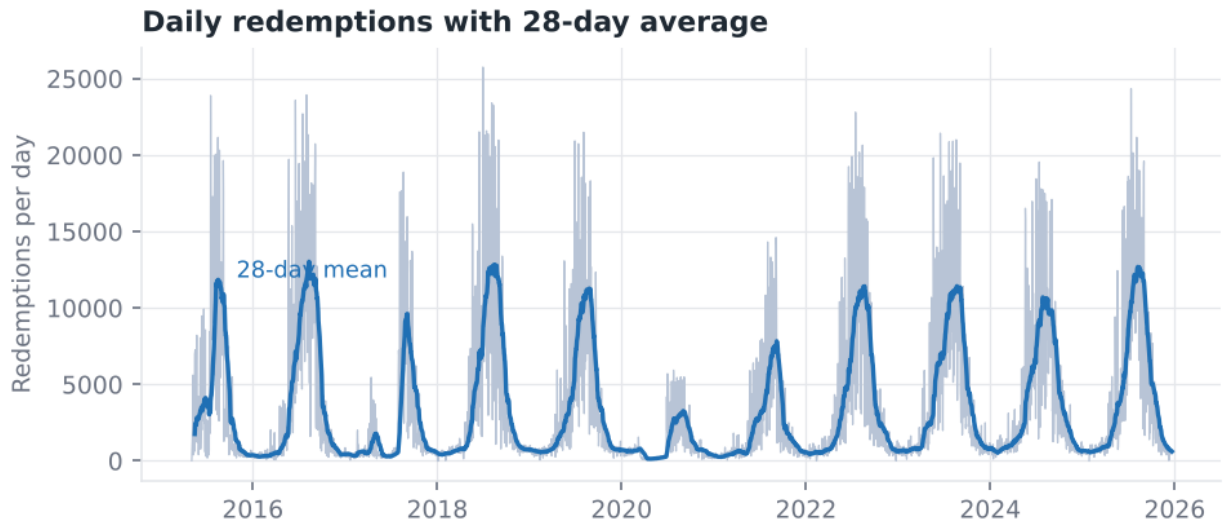


Figure 1. Daily redemptions (grey) with a 28-day average (blue). Summer peaks recur every year; 2017 and 2020 are visibly suppressed.

[Observed] Total redemptions are 12,785,293 and total sales 12,972,051; the mean day has 3,293 redemptions. The busiest day was 2018-07-01 with 25,748 redemptions.

7.2 Hour of day

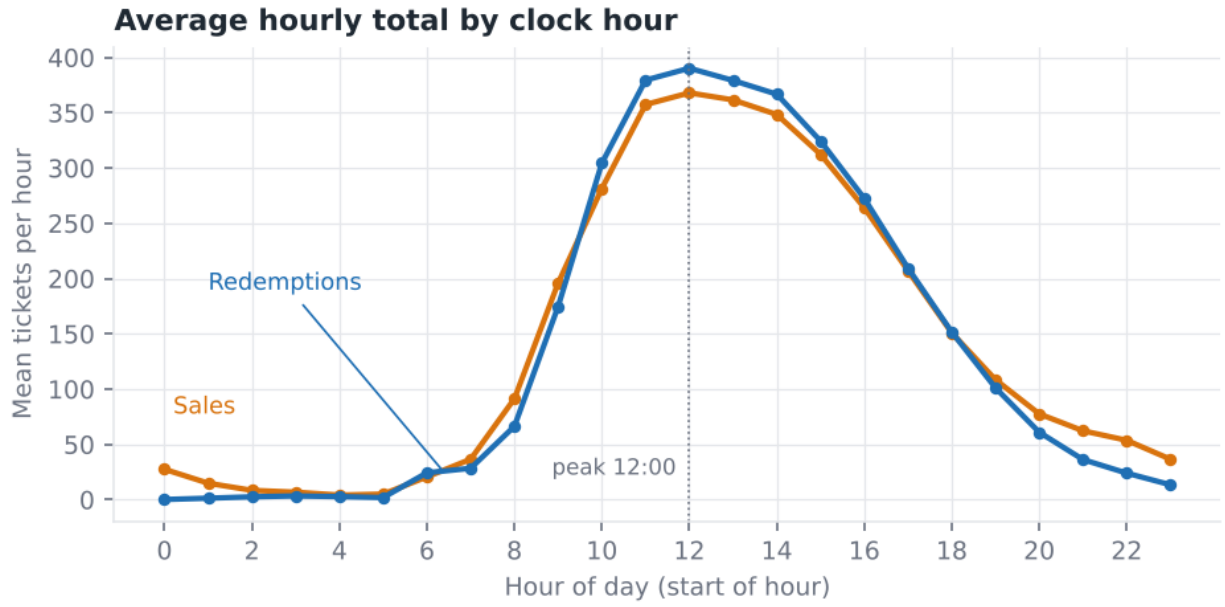


Figure 2. Mean total per clock hour, over days on which the hour has records.

[Observed] Redemptions climb from 28 per hour at 07:00 to a plateau between 11:00 and 14:00 (peak 12:00-13:00, 390 per hour), then decline to 13.5 at 23:00. The hours 10:00 to 15:59 carry 64.9% of all redemptions and 07:00 to 20:59 carry 97.0%. Afternoon and evening (12:00 onward) account for 70% of

redemptions. Sales peak at the same hour (368 per hour) but stay far higher late in the evening and overnight.

[Observed] Overnight behaviour. Between 00:00 and 05:59 there are 15,592 records, yet 98.9% of them show zero redemptions while only 0.75% show zero sales. These hours hold 0.82% of all sales but 0.11% of redemptions. **[Hypothesis]** Sales outside service hours may be advance or online purchases redeemed later; this is consistent with, but not proven by, the data. It is also why sales and redemptions should not be treated as the same event.

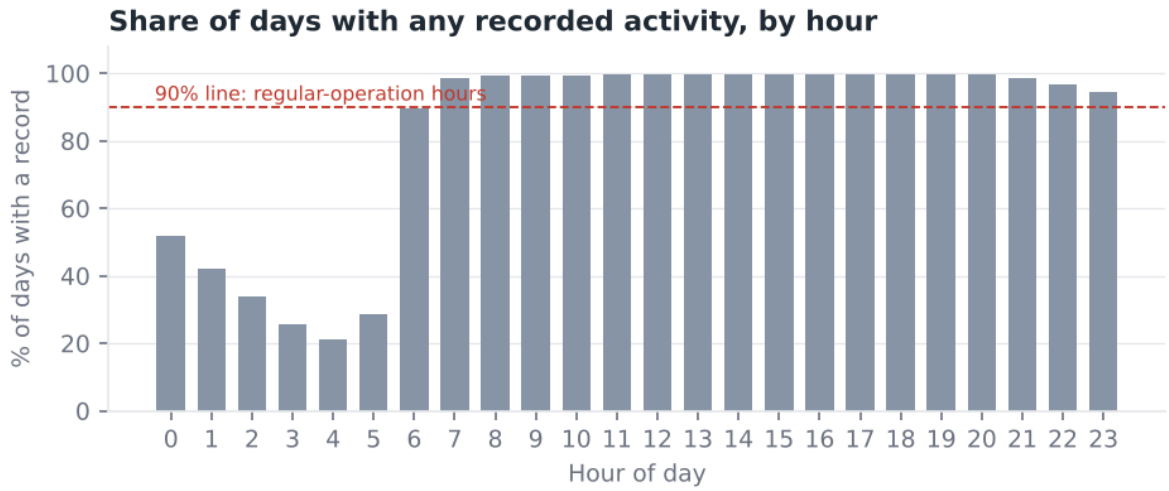


Figure 3. Hours before 07:00 have records on well under 90% of days, which is why they are excluded from off-peak ranking.

7.3 Day of week

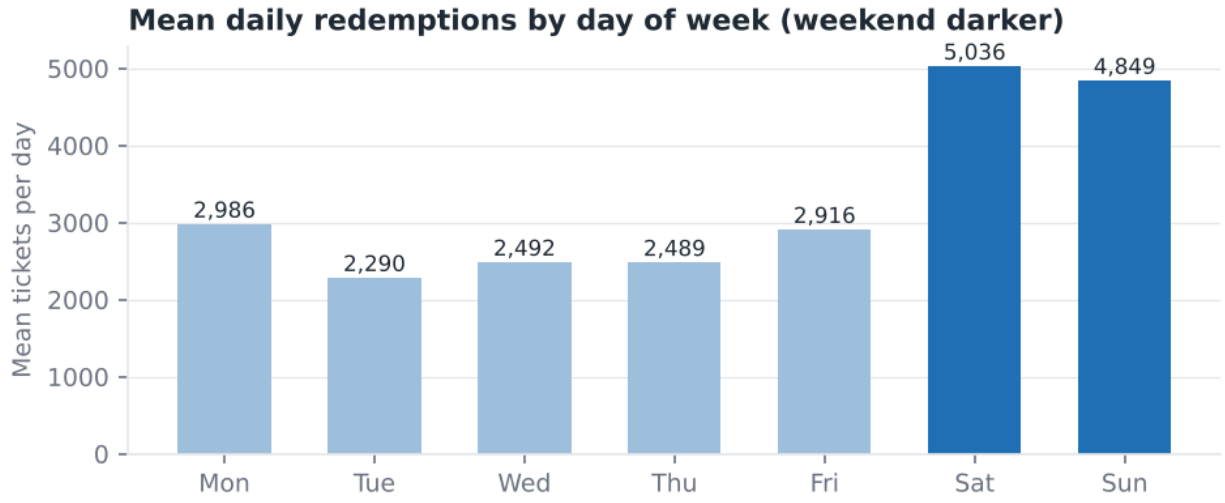


Figure 4. Mean daily redemptions by weekday.

[Observed] Saturday (5,036) and Sunday (4,849) are the busiest days; Tuesday (2,290) is the quietest. Weekends average 1.88 times weekdays (4,943 vs 2,635 per day). Monday and Friday sit above Tuesday to Thursday. **[Observed]** The weekend effect is seasonal: 1.91x in summer but 0.99x in winter. The single busiest weekday-hour is Saturday 13:00 (643 redemptions per hour on average). **[Hypothesis]** Monday and Friday may be lifted by long weekends and holidays; the analysis does not separate holidays.

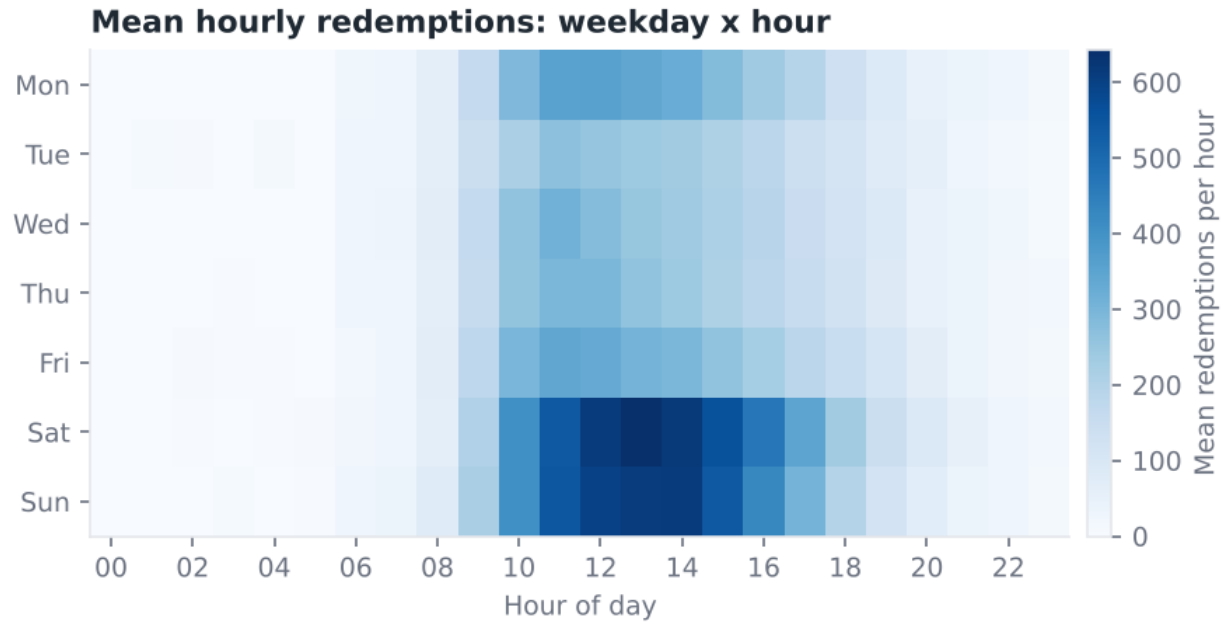


Figure 5. Weekday-by-hour intensity: the busy zone is a weekend, late-morning to mid-afternoon block.

7.4 Month and season

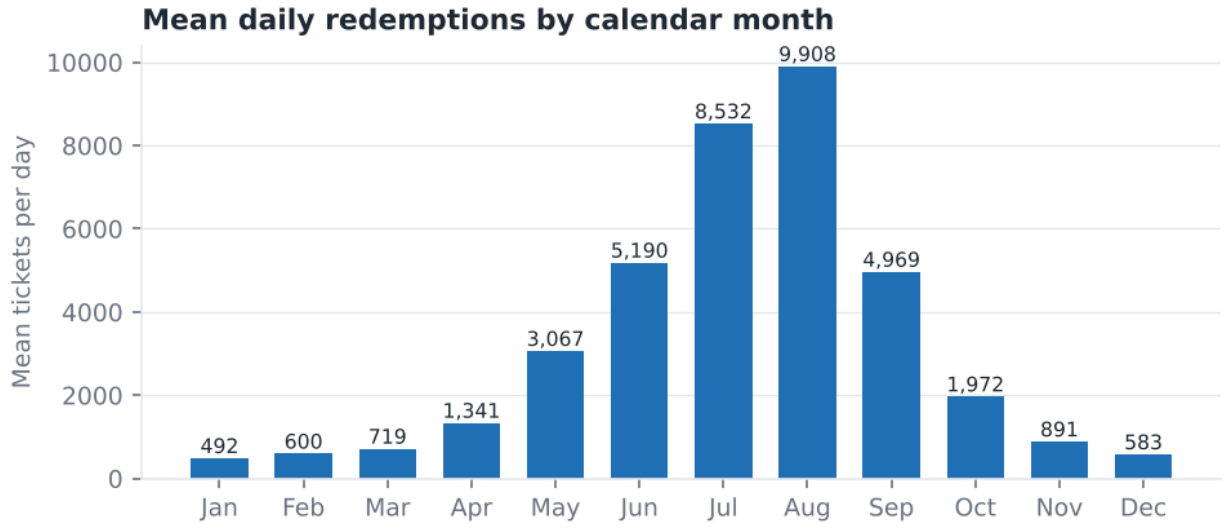


Figure 6. Mean daily redemptions by calendar month, all years pooled.

[Observed] August is the busiest month (9,908 redemptions per day), followed by July (8,532); January is the quietest (492). July and August together hold 49.2% of all redemptions in the file.

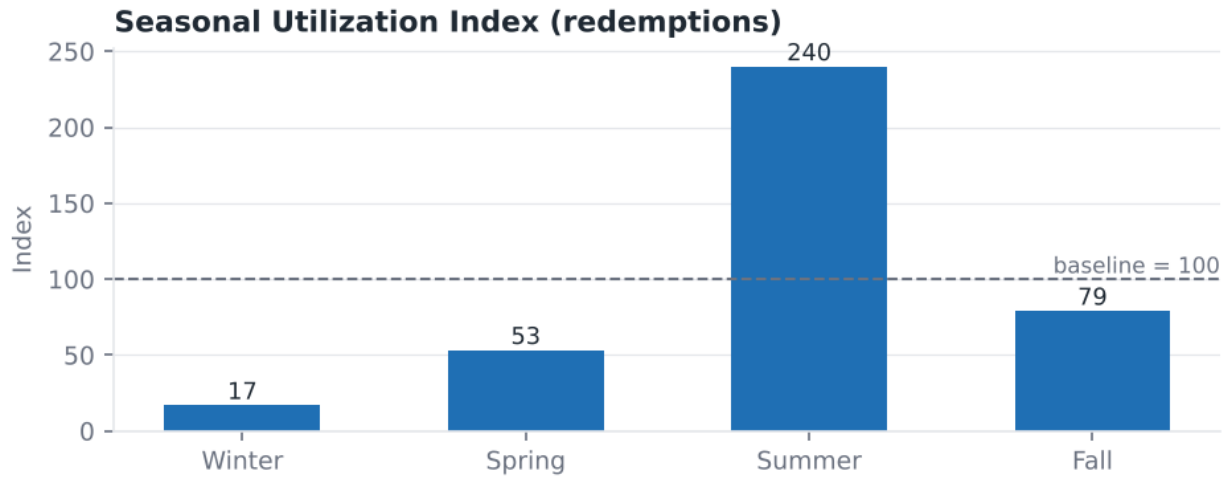


Figure 7. Seasonal Utilization Index; the baseline of 100 is the average day of the whole file.

[Observed] Seasonal index: Winter 17, Spring 53, Summer 240, Fall 79. The low season runs at 7% of the peak season's daily level (Winter versus Summer).

7.5 Year-over-year behaviour and anomalous years

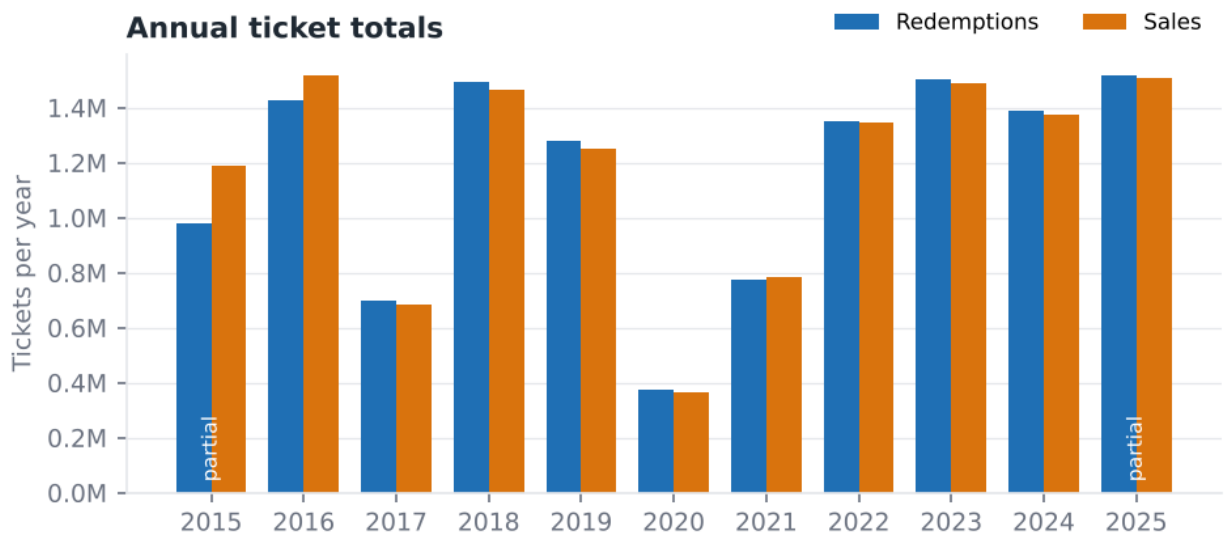


Figure 8. Annual totals. Partial years: 2015 begins on 1 May and 2025 ends on 21 December.

Year	Days with data	Redemptions	Sales	Sales / redemptions
2015	243	978,707	1,189,620	1.216
2016	366	1,425,779	1,518,428	1.065
2017	365	698,861	682,346	0.976
2018	364	1,493,560	1,463,589	0.980
2019	365	1,278,505	1,249,725	0.977
2020	366	374,546	366,606	0.979
2021	363	773,040	782,368	1.012
2022	365	1,349,778	1,346,659	0.998
2023	365	1,502,883	1,491,473	0.992
2024	366	1,390,358	1,373,563	0.988

Year	Days with data	Redemptions	Sales	Sales / redemptions
2025	354	1,519,276	1,507,674	0.992

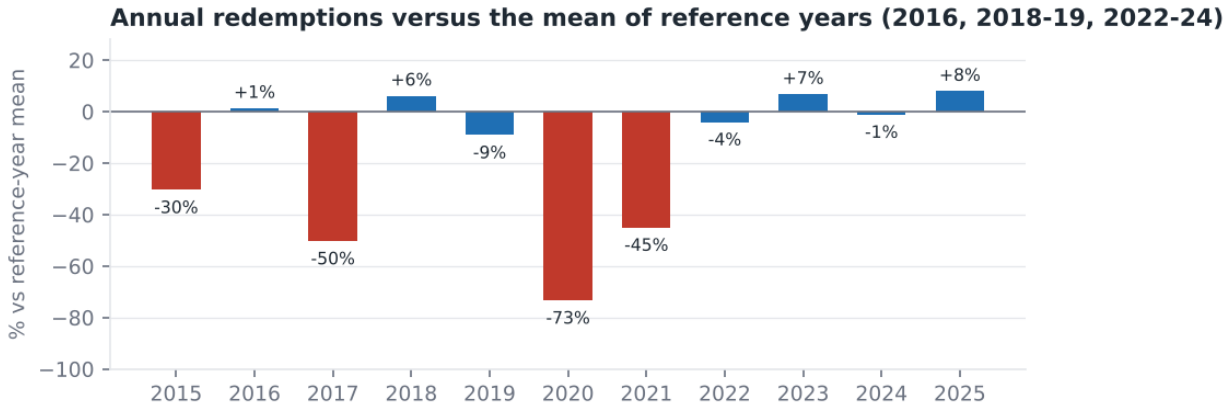


Figure 9. Annual redemptions relative to the mean of six reference years. 2015 and 2025 are partial years and are not comparable on a calendar-year basis.

[Observed] 2017 and 2020 depart sharply from the pattern. Redemptions from May to July 2017 were 45,342, which is 93% below the mean of the same months in the reference years. Redemptions from April to September 2020 were 266,124, 78% below the reference mean for those months. 2021 remained 45% below the reference annual mean, and 2025 (through 21 December) was 8% above it and the highest annual total in the file (1,519,276). **[Hypothesis]** These could reflect service disruptions, closures, or data-capture problems. The dataset contains no cause, and none is asserted here; operations staff should confirm before these years are used as baselines.

7.6 Sales versus redemptions

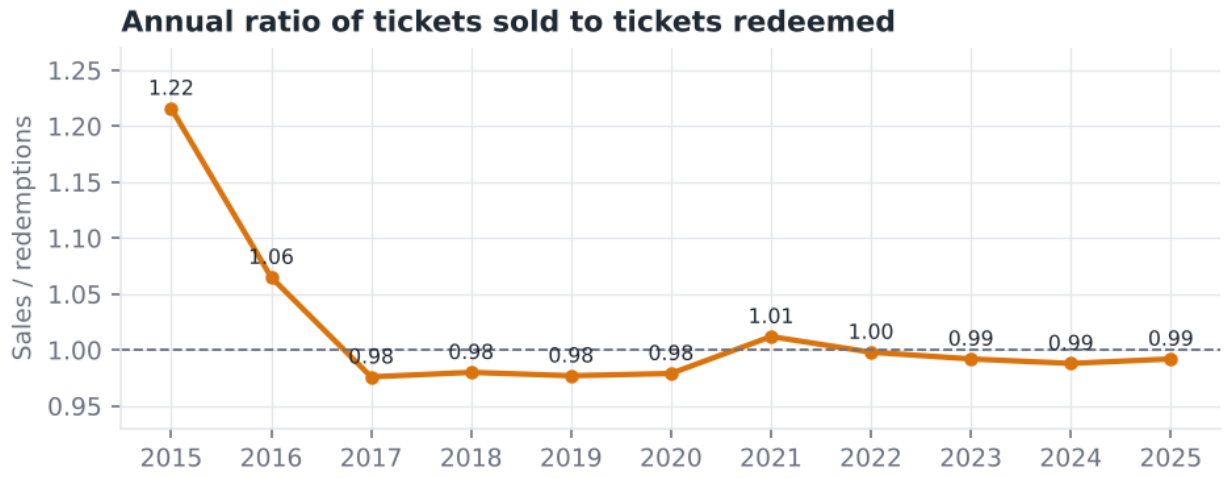


Figure 10. Annual sales-to-redemption ratio.

[Observed] Over the whole file sales exceed redemptions by 186,758 tickets (ratio 1.015). Daily totals are very closely related (correlation 0.981); 15-minute totals less so (0.918). The ratio is not stable: 1.22 in 2015 and 1.06 in 2016, then between 0.98 and 1.01 from 2017 onward. **[Hypothesis]** The early excess may reflect unredeemed tickets or a change in recording or ticket products; the data cannot tell. Either way, the gap is a ticket-count difference and not a measure of people on the water.

7.7 Rolling demand and unusual spikes

[Observed] On a regular 15-minute grid (absent intervals taken as zero), the highest 4-hour trailing mean was 783 redemptions per 15 minutes (about 3,132 per hour) ending 2018-07-08 15:00. The highest 1-hour mean was 1,816 per 15 minutes, ending 2023-08-15 21:00, driven by a single extreme interval.

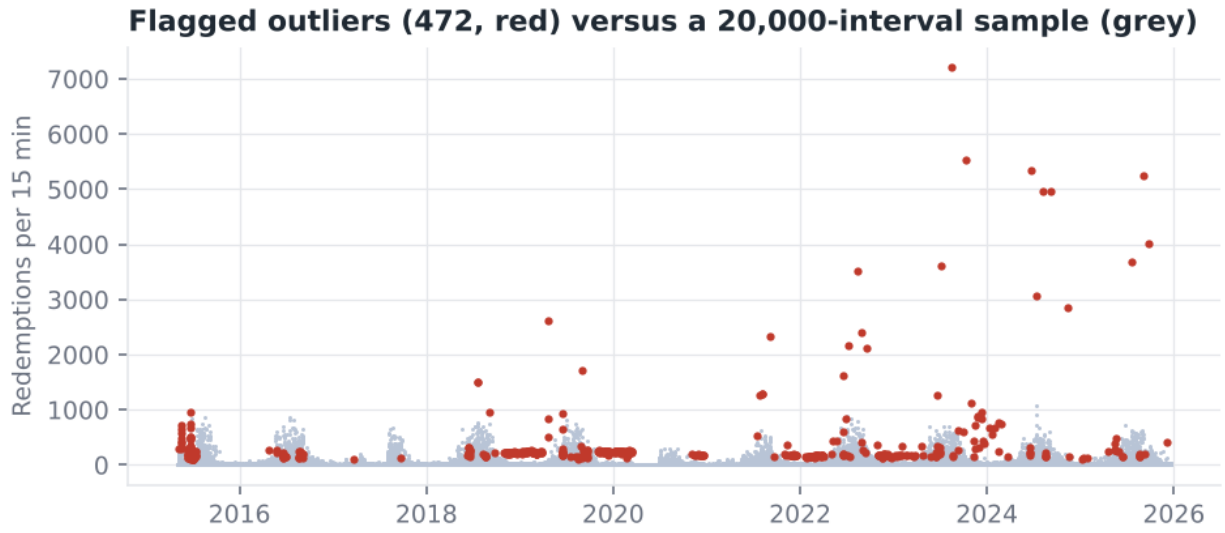


Figure 11. Flagged outliers (red) against a random sample of normal intervals (grey).

[Observed] 472 intervals are flagged; they account for 1.5% of all redemptions. The largest, 7,216 redemptions and 7,229 sales at 2023-08-15 20:15, sits between neighbouring intervals of 4 to 27 redemptions. In each of the eight largest flagged intervals, sales and redemptions agree within 1%. **[Hypothesis]** This pattern is typical of a bulk or group transaction, or of a batch-recording artefact, rather than ordinary walk-up demand. These records are retained because genuine event demand cannot be ruled out, but they should be verified before being used for capacity design.

8. Peak and off-peak demand

Period	Finding	Value
Peak hour	Highest mean hourly redemptions	12:00-13:00, 390 per hour
Peak 3-hour window, all days	Share of average daily redemptions	11:00-14:00, 34.6%
Peak 3-hour window, summer	Same measure, Jun-Aug only	11:00-14:00, 36.8%
Peak 3-hour window, winter	Same measure, Dec-Feb only	14:00-17:00, 28.9%
Peak weekday-hour	Highest mean hourly cell	Saturday 13:00, 643 per hour
Peak month	Highest mean daily redemptions	August, 9,908 per day
Off-peak hour (regular operation)	Lowest mean hourly total, hours active on >=90% of days	23:00-24:00, 13.5 per hour
Off-peak season	Lowest seasonal index	Winter, index 17
Off-peak month	Lowest mean daily redemptions	January, 492 per day
Overnight (00:00-05:59)	Sporadic activity, almost all sales-only	0.11% of redemptions

The winter peak window shifts to the afternoon (14:00-17:00) and is less concentrated (28.9% of the daily total versus 36.8% in summer); the reason is not in the data.

9. Operational implications

Demand is both spiky and seasonal: the average summer day is 2.4 times the overall average day and the average winter day only 0.17 times, and summer weekends run at 1.9 times summer weekdays. Planning that matches the overall average day would therefore under-serve the peak and over-serve the trough. The largest sustained load in the file is about 3,132 redemptions per hour over four hours (July 2018); whether that approaches physical capacity cannot be judged without vessel and queue data, which this dataset does not contain. Because sales continue overnight while redemptions do not, any system that must react to arrivals should be driven by redemptions, not sales.

10. Recommendations supported by the evidence

These are hypotheses for operations to validate; none has been tested against capacity, cost or service obligations.

- **[Recommendation]** Align staffing, ticketing and queue management with the late-morning to mid-afternoon window (11:00-14:00 holds 37% of summer daily redemptions), with the heaviest cover on summer Saturdays and Sundays.
- **[Recommendation]** Plan on three tiers: summer (index 240), shoulder (Spring 53, Fall 79) and winter (17); evaluate whether a reduced winter service pattern is consistent with service obligations, since December to February carries only 4.0% of tickets.
- **[Recommendation]** Apply weekend-specific measures only in the warm season: the weekend premium is 1.91x in summer but 0.99x in winter.
- **[Recommendation]** Build staffing triggers on 4-hour rolling redemption demand rather than single 15-minute values, which are dominated by rare spikes.
- **[Recommendation]** Review the flagged extreme intervals with operations (bulk sales, events, system batching) and document the confirmed cause of the 2017 and 2020 downturns before using those years in baselines or forecasts.
- **[Recommendation]** Capture and share additional data: route or terminal, vessel capacity, queue length or wait time, weather, events, and a real-time feed. These would enable true capacity and real-time analytics, which this file cannot support.

11. Data limitations and assumptions

- Ticket counts are not passenger counts, and sales minus redemptions is not net passenger movement.
- No route, direction, terminal, vessel, capacity, queue, weather or event information is present; no per-destination or capacity conclusion is drawn.
- The file omits intervals with no activity (19.7% of intervals between each day's first and last record are absent). Treating them as zero is an assumption used only for rolling averages.
- Timestamps carry no time zone. They are treated as Toronto wall-clock time on the basis of the DST behaviour described in section 5; this is an inference.
- 2015 starts on 1 May and 2025 ends on 21 December, so annual totals for those years are partial.
- The 2015 and 2016 sales-to-redemption ratios differ from later years for reasons the data do not reveal.
- Analysis is descriptive. No significance tests or forecasts are reported, and the overnight and spike interpretations are hypotheses.
- The dataset's publisher, licence and extraction date are not recorded in the file.

12. Conclusion and future improvements

The supplied dataset is structurally clean and internally consistent, and it supports a clear demand profile: a late-morning to mid-afternoon weekday peak that is amplified on summer weekends, a summer season that carries about 63% of annual tickets, and a winter trough at roughly 7% of summer daily volume. The conclusions most in need of operational validation are the extreme single-interval spikes and the causes of the 2017 and 2020 downturns. Future work should add capacity, queue and weather data; define Sales Count and Redemption Count formally; model demand with holiday and weather effects; and connect the dashboard to a verified live source so that real-time monitoring becomes possible.

Appendix A. Ten busiest days (redemptions)

Rank	Date	Redemptions
1	2018-07-01 (Sun)	25,748
2	2025-07-12 (Sat)	24,341
3	2016-08-01 (Mon)	23,928
4	2015-07-18 (Sat)	23,889
5	2016-06-18 (Sat)	23,584
6	2018-08-05 (Sun)	23,399
7	2018-08-11 (Sat)	23,260
8	2022-07-16 (Sat)	22,798
9	2016-07-17 (Sun)	22,694
10	2016-07-16 (Sat)	22,690

Appendix B. Ten largest 15-minute intervals

Timestamp	Redemptions	Sales	Outlier score
2023-08-15 20:15	7,216	7,229	7.2
2023-10-10 14:00	5,535	5,518	4.7
2024-06-19 19:00	5,341	5,342	4.6
2025-09-01 17:30	5,256	5,218	5.6
2024-08-05 22:00	4,973	4,995	5.3
2024-09-07 17:00	4,966	4,974	5.5
2025-09-24 22:30	4,020	4,015	6.7
2025-07-17 23:15	3,692	3,701	4.4
2023-07-05 21:15	3,602	3,634	5.4
2022-08-15 10:30	3,520	3,408	4.5